

Kriging-Based Spatial Interpolation for Optimizing Device Placement in Low-Coverage Areas

Authors: Howard Yao

Mentors: Stijn Wielandt, Robin Thibaut, Ivo Markovic

Abstract

The next-generation 5G cellular systems, characterized by high throughput and low latency, provide robust support for concepts such as field wireless communication, signal relaying, and remote sensing. Ensuring reliable communication coverage in rugged and remote areas is critical for deploying sensor networks and supporting services. This paper presents an integrated solution that combines a Kriging-based spatial interpolation model with a 5G-enabled portable sensor device developed using Raspberry Pi and a 5G cellular module(RM502Q-AE). The device collects Received Signal Strength Indicator (RSSI) data along with geographical coordinates, including altitude, and transmits this information to a central server in real-time. The Ordinary Kriging algorithm processes the data to estimate signal strength across the terrain, and the results are visualized as GeoTIFF outputs for use in geographical visualization software, like QGIS, and ArcGIS. Our approach accurately identifies optimal device placement and signal relay paths in areas with poor coverage, facilitating more precise exploratory experiments and significantly enhancing network performance and planning. Furthermore, this method is not limited to RSSI analysis but can be applied to the study of any data intensity, offering an efficient and systematic approach to environmental and geological research.

Keywords: 5G, Raspberry Pi, Cellular, GPS, Sensing, Kriging, QGIS

Introduction

Due to complex topography and the lack of existing infrastructure, achieving reliable communication coverage in rugged and remote areas has long been a challenging task. Traditional methods of signal mapping[1] and network planning often fail to account for three-dimensional geographical factors, such as elevation and vegetation density, leading to suboptimal device placement and poor performance. Previous works have employed basic interpolation methods or relied on extensive manual field surveys to estimate signal strength[2]. However, these approaches are either too simplistic or resource-intensive, limiting their

effectiveness in dynamically changing environments. This issue is not confined to network signals but is also relevant to any form of geospatial data analysis.

The primary objective of this project is to develop an integrated solution that combines a custom Kriging-based[3] spatial interpolation model with a 5G-enabled portable sensor device. The device, built using a Raspberry Pi and the RM502Q-AE 5G cellular module, is designed to collect and transmit real-time Received Signal Strength Indicator (RSSI) data along with geographical coordinates, including altitude. The custom Kriging algorithm processes this data to provide accurate signal strength estimates across complex terrains, visualized as GeoTIFF outputs for use in QGIS. This work aims to:

- Accurately map signal strength in remote and rugged environments.
- Identify optimal locations for device placement and signal relay paths.
- Enhance network performance and planning through a systematic and scalable approach.

During the project, additional objectives emerged, such as improving the real-time processing capabilities of the device and expanding the model to include other environmental parameters like vegetation density and atmospheric conditions.

Progress

The technical methodology of this project involves both hardware and software development. At the core of the system is the Raspberry Pi 4 Model B, selected for its balance between performance and portability. It is integrated with an RM502Q-AE 5G cellular module configured with a T-Mobile 5G SIM card, enabling high-speed data transmission even in remote locations. The module can support up to four antenna connections. In the current design, it connects to one GPS antenna for precise geolocation and three cellular signal receiver antennas to ensure reliable cellular connectivity in the field. Additionally, a custom power management system ensures reliable operation during extended field deployments.

The device operates on a lightweight Linux distribution, Raspbian, derived from Debian OS and optimized for the Raspberry Pi. Data collection, processing, and transmission are managed by custom Python scripts. The MQTT protocol is used for reliable and low-latency communication with signal towers at the Lawrence Berkeley National Laboratory (LBNL), enabling real-time data streaming from the field.

Prior to experimentation, the Raspberry Pi's default network manager, which automatically configures available signal sources, must be manually removed. This disables its ability to connect to wireless networks, ensuring that the device relies solely on the 5G module for signal connectivity, thereby avoiding any signal interference. A text-based terminal emulator, Minicom, is used for serial communication with the module, allowing AT command input to monitor and debug the module's status and data transmission, and to conveniently manage the device connected via the serial port.

In practical operation, a Python script on the Raspberry Pi extracts various real-time signal data, including RSSI, through the 5G module, along with real-time GPS coordinates and UTC

timestamps via the GPS antenna. This data is modularized and stored in a CSV file on the Raspberry Pi. The script updates the CSV file at specified time intervals for subsequent analysis. The script is configured as a system service unit file under systemd, a tool used for linux service system management. Allowing it to run automatically on system startup and to restart in case of unexpected termination, such as power issues or 5G module disconnection, ensuring continuous and reliable data collection. As a result, researchers can carry the device, power supply, and radio equipment without connecting to any display or input devices. Simply powering the device initiates automated data processing, ensuring stability and automation in field operations.

Software Development and Data Processing - Kriging

Initially, the collected CSV signal strength data is analyzed using Folium, visualizing data points on a 2D map and representing signal strengths using color maps. However, since the data is discrete and not continuous, this method does not accurately reflect regional conditions. To address this, we utilize the ordinary Kriging algorithm to handle three-dimensional spatial data, incorporating elevation as a critical factor. The collected data undergoes real-time processing to generate and predict missing signal strength in the discrete points, which is then visualized in QGIS.

Kriging is a widely used geostatistical method for spatial interpolation, particularly effective when dealing with regional data that exhibit spatial correlation. The method's core principle is to leverage the spatial relationships between known data points to perform linear, unbiased interpolation for unknown locations. This process begins by constructing a variogram, which quantifies the degree of spatial dependence between data points, assuming that points closer together are more strongly correlated than those farther apart. The variogram reflects how spatial similarity changes with distance. Its quality depends on several factors, such as the spatial distribution and density of data points, which influence the fitting of an optimal curve. Careful adjustment of the variogram is necessary to capture these characteristics accurately, as a well-fitted variogram is essential for producing reliable Kriging results. Therefore, to ensure dataset accuracy, the data is first validated using an older Folium-based Python script on a simple 2-D map with simple points graphed to verify dataset credibility, uniqueness, and eliminate any biased data before running the Kriging algorithm.

For distance and other unique data characteristics, we created a Python script to convert the GPS coordinates recorded in the data to UTM13N format in meters. After determining a suitable variogram, the spatial positions of known points and the variogram are used to calculate the weights of each known point relative to the target point. Kriging aims to assign these weights in a way that minimizes the variance of the interpolated value while ensuring the result remains unbiased. The interpolation is then performed by calculating the expected value at each target point. Initially, a triangular grid was employed to capture finer details in areas with denser data and reduce emphasis on sparsely populated regions. However, to facilitate export as TIFF files for further analysis in QGIS and other GIS software, the grid was ultimately switched to a square format. The choice of grid size directly impacts both the resolution of the spatial variation and the computation time.

Once the accuracy of all data and the brief generated result of Kriging has been verified, a more refined Kriging model can be generated, and the processed data can be formatted into GeoTIFF files for compatibility with QGIS. This enables the results to be overlaid onto real topographic maps, allowing for enhanced visualization and further spatial analysis.

References

- [1] J. Thrane, B. Sliwa, C. Wietfeld and H. L. Christiansen, "Deep Learning-based Signal Strength Prediction Using Geographical Images and Expert Knowledge," GLOBECOM 2020 - 2020 IEEE Global Communications Conference, Taipei, Taiwan, 2020, pp. 1-6, doi: 10.1109/GLOBECOM42002.2020.9322089. keywords: {Predictive models;Radio propagation;Deep learning;Training;Receivers;Computational modeling;Buildings},
- [2] E. Krijestorac, S. Hanna and D. Cabric, "Spatial Signal Strength Prediction using 3D Maps and Deep Learning," ICC 2021 - IEEE International Conference on Communications, Montreal, QC, Canada, 2021, pp. 1-6, doi: 10.1109/ICC42927.2021.9500970. keywords: {Deep learning;Three-dimensional displays;Urban areas;Software algorithms;Stochastic processes;Prediction algorithms;Software;deep learning;UAV;wireless signal strength},
- [3] OLIVER, M. A., & WEBSTER, R. (1990). Kriging: a method of interpolation for geographical information systems. *International Journal of Geographical Information Systems*, 4(3), 313–332. <https://doi.org/10.1080/02693799008941549>